

Personal Statement

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I am third year undergraduate student at Stony Brook University and I am really interested in pursuing a summer research opportunity at Perimeter institute. I hope I can get an opportunity to study quantum many-body theory or quantum gravity in PI with professor Yoshida or professor Gosset, contributing to this fast-developing field and further pursuit a master degree in PSI.

I was found super-intelligent and hard-working at the age of 15 and selected to learn physics at one of the most prodigious universities in China, Jiaotong University. Then, I transferred to Stony Brook University for better resources in both theoretical and experimental high-energy physics in my first year. Previously, in my second year at Stony Brook University, I worked on an atmospheric neutrino reconstruction project for the Deep Underground Neutrino Experiment (DUNE). As the first author, I led the team and substantially improved the neutrino direction reconstruction, energy reconstruction, and particle-antiparticle discrimination based on Monte Carlo simulations. These improvements will lead to better measurement of neutrino oscillation and the CP phase angle. I acquired a lot of quantum simulation and data analysis skills in the process, and the paper will come out within a few months.

I enjoyed the process of working on an experimental high-energy project. Nevertheless, I feel like I should be able to do something more challenging and interesting. During the past three semesters, I started to take upper-level graduate theory courses in order to understand the frontiers of theoretical physics. Thanks to my professors at Stony Brook and YITP, I took two semesters of Quantum Field Theory, Group Theory, and Chern-Simons Theory with Professor Yinchun He from PI. Before the summer, I will further take Quantum Computing and Topological Quantum Field Theory courses to get myself prepared. During the process, I had valuable opportunities to speak with my lecturers, including Professor Zamolodchikov and Professor Nieuwenhuizen, and these conversations strengthened my commitment to pursuing theoretical physics.

Starting in October 2025, I initiated a theoretical research project with Professor Verbaarschot, focusing on quantum chaos in the Sachdev-Ye-Kitaev (SYK) model. The SYK model is a four-Majorana-fermion model first studied in condensed matter physics and later applied to AdS/CFT duality, strange metals, and other areas of theoretical physics. In my analysis, I applied Random Matrix Theory (RMT) to study the model's quantum-chaotic properties, specifically operator growth in out-of-time-ordered correlators (OTOCs) and the spectral density. I argue that the four-local fermion interaction may not be essential and also makes quantum simulation challenging. I introduced several related models with two-local interactions that are analogous to the SYK model, analytically calculated their OTOCs in the large- N limit, and verified the results numerically with finite N . This is significant because our modified models exhibit strong chaotic features without a four-point interaction, providing a promising starting point for quantum simulation and for understanding the original SYK model.

I found this part of Quantum many-body theory very interesting and enthusiastic about the idea of combining computational physics with theoretical foundations in this kind of analysis. Recently I read Professor Yoshida's paper on the evaluation of quantum information scrambling using OTOCs and I find his way of designing quantum circuits and methodology very interesting. Also I think what I am doing right now paves the cornerstone for me to understand his work and further contribute, so I really hope to get an opportunity of collaborating and get supervised under professor Yoshida.

I am also strongly interested in quantum gravity, particularly in approaches that sharpen the connection between quantum information, quantum field theory, and spacetime geometry. Perimeter Institute is especially compelling to me because it hosts a uniquely concentrated community working across complementary directions—ranging from holography and black-hole information to quantum field theory, conformal and topological methods, and the interface with quantum many-body physics. Being in an environment where these perspectives interact daily would allow me to develop a deeper, technically grounded understanding of the field while contributing to concrete research questions at the frontiers of quantum gravity.